

Research Interests

- Developing machine learning methods with a focus on sample difficulty to improve noisy learning.
- Building novel, fair, efficient, and generalizable deep learning models for images, tables, & time-series data.
- Creating AI literacy and governance frameworks and exploring LLMs, VLMs, and agentic AI systems.

Education

University of Toronto, St. George TORONTO, CANADA

Doctor of Philosophy (PhD) – in progress [GPA: A] May 2021 – Present

Computer Science: Machine Learning, Computer Vision (Vector Institute for Artificial Intelligence, The Hospital for Sick Children, Toronto Rehabilitation Institute, UHN KITE)

Courses: Software Engineering for ML, AI Alignment, Ethics of AI in Context, Probabilistic Learning

Supervisors: Dr. Anna Goldenberg, Dr. Babak Taati, Committee Chair: Dr. Rahul Krishnan

Master of Science (MSc) [GPA: A] Sep 2019 – Apr 2021

Computer Science: Machine Learning, Computer Vision (Toronto Rehabilitation Institute, UHN KITE)

Courses: AI & Ethics, Numerical Methods for Optimization, ML for Health, Algorithms for Privacy

Supervisor: Dr. Babak Taati

Trinity College in the University of Toronto, St. George TORONTO, CANADA

Honours Bachelor of Science (HBS) [GPA: 3.81] Sep 2015 – Apr 2019

Dean's List Scholar 2016, 2017, 2018, 2019 – Graduated with High Distinction

Computer Science Specialist and Mathematics Major (Trinity College)

Specialist Focuses: Artificial Intelligence, Scientific Computing, and Theory of Computation

University of Toronto, Mississauga MISSISSAUGA, CANADA

Gifted Student Program [GPA: 4.00] May 2013 – Aug 2014

Took credited Computer Science and Mathematics courses part-time while attending high school

Experience

Faculty of Information, University of Toronto, St. George TORONTO, CANADA

Towards Socially Grounded AI Safety

Research Fellowship Aug 2026 - Present

- Research Fellowship to help build more socially grounded and pluralistically aligned LLMs.
- Conduct literature reviews, assist with writing, build code/prototypes/data, and perform analysis

Supervisors: Dr. Matt Ratto, Dr. Daniel Silver

CoordCare TORONTO, CANADA

Chief Technology Officer

Startup through UTEST Sep 2025 - Present

- Build prototypes for agentic AI systems in healthcare and work with founders to secure investment

Data Sciences Institute, University of Toronto, St. George TORONTO, CANADA

Lecturer/Technical Facilitator

DSI Certificate Program: Git, Unix, and Shell, Winter 2025 Jan 2025

- Held office hours, organized lectures, and marked assignments
- Managed TAs/Learning Support to support students

Department of Computer Science, University of Toronto, St. George TORONTO, CANADA

Research Assistant – Vector Institute

May 2021 – Present

Learning with Noisy Labels

- Developed lightweight methods to learn sample difficulty and improve noisy learning and fairness
- Working on methods for label correction, time-series missingness, and vision-language reasoning

Supervisors: Dr. Anna Goldenberg, Dr. Babak Taati

Research Assistant – UHN KITE Research Institute

Sep 2019 – Present

Automated Facial Pain Detection for Older Adults with Dementia

- Trained an automated facial pain detection algorithm using a pairwise, contrastive method
- Worked with a team at the University of Regina to install, debug, and maintain pain monitoring systems in long-term care homes in Regina and Saskatoon

Supervisor: Dr. Babak Taati

Collaborator: Dr. Thomas Hadjstavropoulos

Research Assistant – The Hospital for Sick Children & The Vector Institute Apr 2019 – Present
Pediatric Whole-Body Cancer Surveillance using GANs & Clinical Predictions from Pediatric Renal Ultrasounds
• Used generative, classification, and segmentation models to aid in pediatric cancer surveillance
• Worked with radiologists to evaluate the efficacy of our AI tools we developed
Supervisor: Dr. Anna Goldenberg Collaborator: Dr. Andrea Doria

Research Assistant – St. Michael's Hospital & Scientific Computing Group Apr 2018 – Apr 2019
Early Stage Lung Cancer Detection: Training a DCNN with Patches from Synthetic Frontal Chest Radiographs
• Generated realistic synthetic nodules and frontal chest radiographs from chest CT scans
• Trained a DCNN to detect early stage lung nodules in a given patch
Supervisor: Dr. Ken Jackson Collaborator: Dr. Joseph Barfett

Teaching Assistant – Department of Computer Science, University of Toronto
CSC110 Foundations of Computer Science I, Fall 2023 Sep 2023 – Dec 2023
• Held tutorials and marked assignments and final exams
Lecturer: Profs. Tom Fairgrieve, Sadia Sharmin
CSC413/2516 Neural Networks and Deep Learning, Winter 2023 Jan 2023 – Apr 2023
• Created and marked assignments and exams, held tutorials, and answered questions on the forum
Lecturer: Profs. Bo Wang, Jimmy Ba
CSC420 Introduction to Image Understanding, Fall 2021 Sep 2021 – Dec 2021
• Held tutorials, ran office hours, answered questions on the forum, and marked final exams
Lecturer: Prof. Babak Taati
CSC165 Mathematical Expression and Reasoning for Computer Science, Summer 2021 May 2021 – Aug 2021
• Marked problem sets and term tests
Lecturer: Prof. Lindsey Shorser
CSC207 Software Design, Winter 2019 Jan 2019 – Apr 2019
• Held two tutorials, ran office hours, marked final presentations and exams
Lecturer: Prof. Lindsey Shorser

Massey College, University of Toronto, St. George TORONTO, CANADA
Junior Fellow Sep 2023 – Present

- Part of the graduate college as a non-resident Junior Fellow
- Co-chaired several committees and participated in the Massey Grand Rounds
- Organized Massey AI Dialogues & Debates and a friend of the Robertson Davies Library
- Helping organize the Couchiching Conference
- The college provides a unique, congenial, and intellectual environment for graduate students of distinguished ability in all disciplines to share in a rich and stimulating community

Temerty Centre for Artificial Intelligence Research and Education in Medicine TORONTO, CANADA
Education Trainee Co-Lead May 2023 – Present

- Guide T-CAIREM's education and training initiatives
- Help organize, moderate, and adjudicate lectures and conferences
- Part of the committee to hire Education team trainees, summer students, and trainees for rounds
- Build capacity in AI in Medicine for high school, undergrad, grad, and post-grad trainees
- Develop & maintain funding opportunities for trainees and organize educational outreach initiatives
- Run AI training and appraisal sessions for medical residents
- Led work on AI governance and AI literacy for TAHSN
- Part of the International AI in Medicine Education Working Group, Data Governance Committee, ACAIM/PCAIM, and the TAHSN Community of Practice for AI in Healthcare
- Co-Lead of Living Glossary in AI and Medicine Subgroup & Medical Resident Training Subgroup

Google MOUNTAIN VIEW, U.S.A.
Google Trailblazer Fall 2013

Raised awareness of greater access to CS education for all (1st place in Hour of Code outreach contest)

Computing and Programming Experience (CAPE) Virtual Summer 2013
Learned about combining design thinking concepts with Android development to solve a social issue

C++ iD Tech Camps Summer 2012
Attended a Google-sponsored hands-on summer technology course to learn C++ at Stanford University

Computing and Programming Experience (CAPE) Summer 2011
Experienced daily life at Google and learned about algorithms, systems, programming, and computing

In Submission

- Moturu, A.***, Postill, G.*, Rosella, L., Mamdani, M. (2026). A Pan-Canadian Plan for AI Education. Submitting to *Nature Communications, AI in Education*.
- Moturu, A.***, Postill, G.*, Rosella, L., Mamdani, M. (2026). Towards an Healthcare AI Governance Framework. Submitting to *npj Digital Medicine, Exploring the impact of governance models in AI and digital health on healthcare*.
- Postill, G.*, **Moturu, A.***, Rosella, L., Mamdani, M. (2026). Towards an Healthcare AI Literacy Framework. Submitting to *npj Digital Medicine, Transforming Medical Education through Artificial Intelligence*.
- Moturu, A.**, Muzammil, M., Goldenberg, A., Taati, B. (2026). **LiLAW: Lightweight Learnable Adaptive Weighting to Learn Sample Difficulty & Improve Noisy Training**. In submission at *IEEE Transactions on Pattern Analysis and Machine Intelligence*.
- Moturu, A.**, Goldenberg, A., Taati, B. (2026). **LiBaGS: Lightweight Boundary Gap Synthesis for Targeted Synthetic Data Selection**. In submission at the *Association for the Advancement of Artificial Intelligence Conference on Artificial Intelligence*.
- Moturu, A.**, Taati, B.*, Goldenberg, A.* (2026). **LiNC: Lightweight Noise Correction via Per-Sample Trust and Gaussian Mixture Modeling**. In submission at the *Association for the Advancement of Artificial Intelligence Conference on Artificial Intelligence*.
- Jeddi, A., Shaban, K., Baghbanzadeh, N., Sharan, N., **Moturu, A.**, Taati, B. (2026). **When Does RL Help Medical VLMs? Disentangling Vision, SFT, and RL Gains**. In submission.
- Bestavros, S., Chen, E., **Moturu, A.**, Rahman, S., Tyli, R., Wang, S. (2026). A Narrative Review of Artificial Intelligence-Based Imaging in Dermatology: From Potential to Practice. In submission at the Canadian Dermatology Association Conference.
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Publications

- Moturu, A.**, Komolafe, R., Joshi, S., Doria, A., Goldenberg, A. (2025). **Human-AI Interaction With AI-Assisted Tumor Overlays in Pediatric Whole-Body Magnetic Resonance Imaging: Exploratory Reader Study**. In *JMIR Human Factors*.
- Postill, G., Midroni, J., **Moturu, A.**, Rosella, L. and Haque, N. (2026). **Moving beyond one-size-fits-all education approaches for artificial intelligence in healthcare**. In *PLOS Digital Health*, 5(5), p.e0001408.
- Midroni, J., Postill, G., **Moturu, A.**, Haque, N., Rosella, L. (2026). **Building an Interdisciplinary Centre for AI Education: Strategy and Practice**. In *University of Toronto Medical Journal* 103.
- Ratto, M., **Moturu, A.**, Silver, D. (2026). **Socially Grounded Agentic AI: Coordinating Plural Perspectives through Social Theory**. In *Pluralistic Alignment Workshop at the International Conference on Machine Learning*.
- Lin, L., Mehraban, S., **Moturu, A.**, Taati, B. (2026). **Pain in 3D: Generating Controllable Synthetic Faces for Automated Pain Assessment**. In the *International Conference on Pattern Recognition*.
- Taati, B., Muzammil, M., Zarghami, Y., **Moturu, A.**, Kazerouni, A., Reimer, H., Mihailidis, A., Hadjistavropoulos, T. (2025). **SynPAIN: A Synthetic Dataset of Pain and Non-Pain Facial Expressions**. *arXiv preprint arXiv:2507.19673*.
- Stopyn, R. J. N., **Moturu, A.**, Taati, B., Hadjistavropoulos, T. (2025). **Real-Time Evaluation of an Automated Computer Vision System to Monitor Pain Behaviour in Older Adults**. In *Journal of Rehabilitation and Assistive Technologies Engineering*.
- Postill, G., **Moturu, A.** (2024). Integrating AI into Medical Education: A Curriculum for Appraising AI Studies in Clinical Practice. Workshop presented at *International Congress on Academic Medicine*, Vancouver, BC.
- Moturu, A.**, Joshi, S., Doria, A., Goldenberg, A. (2022). **Volume-based Performance not Guaranteed by Promising Patch-based Results in Medical Imaging**. (Spotlight) In *Proceedings on "I Can't Believe It's Not Better! – Understanding Deep Learning Through Empirical Falsification"* NeurIPS Workshop, PMLR 187:85-93.
- Rezaei, S., **Moturu, A.**, Zhao, S., Prkachin, K. M., Hadjistavropoulos, T., Taati, B. (2020). **Unobtrusive Pain Monitoring in Older Adults with Dementia using Pairwise and Contrastive Training**. In *IEEE Journal of Biomedical and Health Informatics*, doi: 10.1109/JBHI.2020.3045743.
- Rezaei, S., **Moturu, A.**, Zhao, S., Prkachin, K. M., Hadjistavropoulos, T., Taati, B. (2020). **Ambient Pain Monitoring in Older Adults with Dementia to Improve Pain Management in Long-Term Care Facilities**. In *ACM International Conference on Multimodal Interaction 2020, Multimodal Interaction in Psychopathology*.
- Chang, A.*, Suriyakumar, V. M.*, **Moturu, A.***, Tewattanarat, N., Doria, A., Goldenberg, A. (2020). **Using Generative Models for Pediatric wbMRI**. In *Medical Imaging with Deep Learning*.
- Moturu, A.**, Chang, A. (2018). **Creation of Synthetic X-Rays to Train a Neural Network to Detect Lung Cancer**. In *Review of Undergraduate Computer Science, University of Toronto*.

Reports & Preprints

- Rosella, L. et al. (2025). [Global Framework for an AI Curriculum in Medicine](#). International AI in Medicine Education Working Group.
- Rosella, L. et al. (2025). [Living Glossary for AI and Medicine](#). International AI in Medicine Education Working Group.
- Dubber, M. D., Pasquale, F., Das, S. (Eds.). **Moturu, A.** et al. (2022). [The Oxford Handbook of Ethics of AI: An Annotated Bibliography](#). In *Oxford Handbooks*.
- Chang, A., Suriyakumar, V. M., **Moturu, A.**, Tu, J., Tewattananat, N., Joshi, S., Doria, A., Goldenberg, A. (2021). [3D Reasoning for Unsupervised Anomaly Detection in Pediatric WbMRI](#). *arXiv preprint 2103.13497*.
- Moturu, A.**, Sabo, A., Jafari, D., Korhani, N. (2020). [Deep Learning for View Labelling of Pediatric Renal Ultrasounds for Predicting Surgical Intervention, Vesicoureteral Reflux, and Kidney Function](#). CSC2541H1S Machine Learning for Healthcare course project.
- Chang, A., **Moturu, A.** (2019). [Detecting Early Stage Lung Cancer using a Neural Network Trained with Patches from Synthetically Generated X-Rays](#). In *Numerical Analysis Technical Reports, University of Toronto*.
- Moturu, A.**, Chang, A. (2018). [Creation of Synthetic X-Rays to Train a Neural Network to Detect Lung Cancer](#). In *Numerical Analysis Technical Reports, University of Toronto*.
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Organizing & Reviewing

- 2026 CIC's Couchiching Conference (**Organizer**)
Neural Information Processing Systems
Machine Learning for Healthcare
- 2025 IEEE / CVF Computer Vision and Pattern Recognition Conference
T-CAIREM Conference: The Evolution of Generative A.I. (**Organizer**)
Machine Learning for Healthcare
International Conference on Machine Learning
- 2024 International Conference on Learning Representations: Time Series for Healthcare Workso
T-CAIREM Symposium: Multi-Modal Data and the Future of Health AI (**Organizer**)
- 2023 T-CAIREM Conference: AI in Medicine (**Organizer**)
International Conference on Medical Image Computing and Computer Assisted Intervention
MICCAI Workshop: AmI4HC
- 2022 NeurIPS Workshop: Learning from Time Series for Health
Machine Learning for Healthcare
- 2021 Machine Learning for Healthcare
IEEE Transactions on Affective Computing
- 2020 Machine Learning for Healthcare
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Talks

- 2026 International Conference on Machine Learning @ Seoul, South Korea (**Presenter**)
Emerging Leaders in Healthcare Conference @ Cal Poly San Luis Obispo (**Presenter**)
MaLMIC Open Forum on Validating AI in Healthcare: Generalizability and Explainability (**Presenter**)
Artificial Intelligence High Table with Dr. Geoffrey Hinton @ Massey College (**Invited Guest**)
(Eh)I: Artificial Intelligence in the Canadian Context with Dr. Geoffrey Hinton @ Massey College
CIC Bill Graham Lecture with Dr. Geoffrey Hinton @ The Arcadian
- 2025 Academic Half Day: Psychiatry and AI @ T-CAIREM (**Presenter**)
Hinton Lectures @ Metro Toronto Convention Centre
T-CAIREM AI in Medicine Conference @ Marriott, Toronto (**Adjudicator**)
Academic Half Day: Neurology and AI @ T-CAIREM (**Presenter**)
Massey Grand Rounds with Dr. James Orbinski @ Massey College
- 2024 AGE-WELL Conference @ Delta, Edmonton (**Presenter**)
Doctoral Symposium: Conference on Health, Inference, and Learning @ Cornell Tech, NYC (**Presenter**)
T-CAIREM Symposium: Multi-Modal Data and the Future of Health AI @ Hart House
Artificial Intelligence Academic Dinner @ Trinity College (**Organizer**)
- 2023 Academic Half Day: Cardiac Surgery and AI @ T-CAIREM (**Presenter**)
Massey Dialogues on AI and Healthcare: Challenges & Opportunities @ Massey College (**Organizer**)
T-CAIREM AI in Medicine Conference @ Metro Toronto Convention Centre (**Organizer and Presenter**)
Massey Grand Rounds with Dr. Janet Rossant @ Massey College

	AI4Health Summer School @ PariSanté Campus, Paris (Presenter)
	Machine Learning High Table @ Trinity College (Organizer)
2022	ICBINB Workshop Presentation @ NeurIPS in New Orleans, LA, USA (Spotlight Presenter)
	AI in Healthcare with Drs. Babak Taati, Frank Rudzicz, Farzad Khalvati @ Trinity College (Organizer)
2021	Dr. Jeffrey Cohn, Multimodal Measurement of Depression and Treatment-Resistant OCD @ UHN KITE
	Computer Vision Journal Club: Automatic Pain Recognition @ UHN KITE (Presenter)
2019	Dr. Geoffrey Hinton's Turing Lecture & Symposium @ the Sheraton Centre Toronto Hotel (Presenter)
	BSI Remote International Conference @ the University of Toronto (Best Presentation Award)
	Trinity College Undergraduate Research Conference @ Trinity College, University of Toronto (Presenter)

Awards

2026	Bell Graduate Scholarship	\$20,000
	Massey College OSOTF – HUDD Bursary	\$1,500
	Monica Ryckman Bursary	\$4,000
	Vector Institute Research Scholarship	\$6,000
	Massey College Junior Fellowship	
2025	Frederick Hudd Scholarship – Massey College	\$2,000
	T-CAIREM Student Education Co-Lead	\$4,000
	Ontario Graduate Scholarship	\$15,000
	Vector Institute Research Scholarship	\$6,000
	Massey College Junior Fellowship	
2024	Frederick Hudd Scholarship – Massey College	\$3,000
	Ontario Graduate Scholarship	\$15,000
	Vector Institute Research Scholarship	\$6,000
	T-CAIREM Student Education Co-Lead	\$4,000
	Massey College Junior Fellowship	
2023	Ontario Graduate Scholarship	\$15,000
	Vector Institute Research Scholarship	\$6,000
	T-CAIREM Student Education Co-Lead	\$4,000
	Massey College Junior Fellowship	
2022	Chancellor William C. Graham Award - Trinity College	
	Vector Institute Research Scholarship	\$6,000
2021	Queen Elizabeth II Graduate Scholarship in Science & Technology	\$15,000
2020	CIHR Frederick Banting and Charles Best Canada Graduate Scholarship-Master's (CGS-M)	\$17,500
2019	NSERC Undergraduate Student Research Award (USRA) [Supervisor: Dr. Anna Goldenberg]	\$6,000
	Platterz Prize in Computer Science	\$2,500
	BSI Remote International Conference Best Presentation Award	\$100
2018	Trinity College Drew Thompson Scholarship	\$300
	NSERC Undergraduate Student Research Award (USRA) [Supervisor: Dr. Ken Jackson]	\$6,000
2016	Trinity College Chancellor's Scholarship – Frederick K. Ashbaugh	\$200
2015	President's Scholars of Excellence Program [Mentor: Dr. Gerald Penn]	\$10,000
	C. David Naylor University Scholarship	\$20,000

Media

- Samsel, Konrad. T-CAIREM co-designs a health AI workshop series for patients involved in medical research, *T-CAIREM*, 22 Jul. 2026, <https://tcairem.utoronto.ca/news/t-cairem-co-designs-health-ai-workshop-series-patients-involved-medical-research>.
- Temerty Centre for Artificial Intelligence Research and Education in Medicine. "Meet Gemma Postill and Abhishek Moturu, T-CAIREM's New Student Education Co-Leads", *T-CAIREM*, 18 Apr. 2023, <https://tcairem.utoronto.ca/news/meet-gemma-postill-and-abhishek-moturu-t-cairems-new-student-education-co-leads>.
- Chancellor William C. Graham Awards. "Chancellor William C. Graham Awards 2022", *Trinity College*, 31 Oct. 2022, <https://www.trinity.utoronto.ca/discover/news/item/congratulations-to-the-recipients-of-the-2022-chancellor-william-c-graham-awards/>.
- Perez, Ryan. "From Data Visualization to AI in Health Care: Big Ideas, New Solutions from U of T Undergraduate Researchers." *University of Toronto News*, 13 Sept. 2018, <https://www.utoronto.ca/news/data-visualization-ai-health-care-big-ideas-new-solutions-u-t-undergraduate-researchers>.

Volunteering

Trinity College in the University of Toronto, St. George

TORONTO, ONTARIO, CANADA

- Year Representative: Alumni Reunion Planner Feb 2024 – Present
- Fundraiser and Donor: Trinity College Donald Rickerd Scholarship Nov 2025 – Present
- Fundraiser and Donor: Artificial Intelligence and Society Scholarship Nov 2025 – Present
- Donor: Viridis Project Jan 2025 – Present
- Fundraiser and Donor: Sylvia Lassam Scholarship Jan 2025 – Dec 2025
- Refreshments Coordinator: Conversazione Ball Jan 2019 – Apr 2019
- Social Media Manager: Environmental Society Sep 2018 – Apr 2019
- Commuter Coordinator, App Developer, Parade Driver: Orientation Week Sep 2016, 2017, 2018
- Logistics Coordinator: Saints Ball Sep 2017 – Dec 2017
- Sponsorship Coordinator: Conversazione Ball Jan 2017 – Apr 2017

Massey College, University of Toronto, St. George

TORONTO, ONTARIO, CANADA

- Fundraiser and Donor: Massey College Donald Rickerd Scholarship Nov 2025 – Present
- Co-chair: Lionel Massey Fund and House Committee Sep 2023 – Aug 2024
- Co-chair: Computing Committee and Junior Fellow Lecture Series Committee Sep 2023 – Aug 2024

Department of Computer Science, University of Toronto, St. George

TORONTO, ONTARIO, CANADA

- Ukraine Undergraduate Summer Exchange Program Jun 2022 – Aug 2022
- Toronto Graduate Application Assistance Program Oct 2021 – Nov 2021
- University of Toronto Artificial Intelligence Team ProjectX Domain Expert (ML Research Competition focused on Climate Change and Infectious Diseases) Sep 2020 – Apr 2021
- University of Toronto Machine Intelligence Student Team Project Director (ML Project on Brain Tumour Segmentation and Survival Prediction) Sep 2019 – Apr 2020
- Department of Computer Science (DCS) Ambassador Sep 2017 – Jun 2019
- Undergraduate Theory Group (UTG) VP Communications Sep 2017 – Apr 2018

Department of Mathematics, University of Toronto, St. George

TORONTO, ONTARIO, CANADA

- Math Union VP Communications and Secretary Sep 2017 – Apr 2019
- Department of Mathematics Undergraduate Committee Member Sep 2017 – Apr 2018
- Department of Mathematics Peer Mentor Sep 2016 – Apr 2017

Affiliations

University of Toronto, St. George	2015 – Present
Trinity College	2015 – Present
The Hospital for Sick Children	2019 – Present
UHN KITE Research Institute	2019 – Present
Vector Institute for Artificial Intelligence	2021 – Present
Massey College	2023 – Present
Temerty Centre for Artificial Intelligence Research and Education in Medicine	2023 – Present
Faculty Club	2024 – Present
Data Sciences Institute	2025 – Present
Canadian International Council	2025 – Present
Faculty of Information	2026 – Present
University of Toronto, Mississauga	2013 – 2014
Google	2011 – 2013

Skills

Areas

Machine Learning, Computer Vision, Deep Learning, Natural Language Processing, Numerical Methods, Data Structures, Algorithm Design, Complexity, Probability, Calculus, Algebra, Logic

Programming Languages

Python, Julia, MATLAB, Java, C, C++, Swift, LaTeX

Tools & Libraries

PyTorch, TensorFlow, OpenCV, Numpy, Scipy, Scikit-Learn, NLTK, Matplotlib, Git, Android, Linux

Languages

English (*native proficiency*); French, Italian, Hindi (*elementary proficiency*); Telugu (*native proficiency*)